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1. SFR's have several inherent safety benefits. These include a large thermal response time, high boiling point of the coolant, and a generally negative reactivity feedback coefficient due to thermal expansion of core components leading to additional leakage. The boiling point of sodium is high leading to a lot of margin between operating conditions and significant reduction in cooling due to vapor coolant. Finally, sodium melts at 98°C so high pressure RCS is not required for SFRs.

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2. Uranium is alloyed often with Zr for several reasons including stabilizing the gamma phase, increasing the melt temperature, reducing anisotropic swelling, and improving mechanical properties. Of these, reducing anisotropic swelling is essential as unalloyed alpha phase U experiences significant anisotropic swelling due to interstitial and void seeming along with anisotropic thermal expansion. Combined with anisotropic swelling in grains, the differential strains lead to differential stresses which are relieved through plastic strain (texturing or cavitation swelling). This is the main method of anisotropic swelling at up to 600°C. Alloying with other metals form precipitates for point defect sinks and lower the anisotropic gamma phase boundary.

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3. Metallic fuel systems undergo significant radial swelling. Much of this is contributed to the differential stresses due to high plasticity gamma U-Pu-Zr developing large fission gas bubbles while low plasticity alpha U-Pu-Zr (low temp region) resists expansion. This leads to a higher radial stress and subsequent radial strain. To alleviate this problem, a low smear density (75%) is needed to allow around 30% swelling of the fuel before PCMI. At that point, the fuel typically has enough open porosity that FG release alleviates the swelling. The plenum accomodates high quantities of released fission gas (1.5 times as long as fuel stack) preventing rupture due to high rod internal pressure.

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4. Constituent redistribution is the creation of Zr enriched central zones, Zr depleted - Uranium enriched mid zones, and as-fabricated zones in the fuel. Methods of transportation are two-fold, with thermo-dynamic gradients and chemical potential brought to contribute. The Soret effect plays a role where the Soret coefficient of Zr is -34 kcal/mol while U is 5 kcal/mol. Therefore, Zr tends to move up the temperature gradient while U moves down. The effects here include additional swelling and different microstructures and porosity. The porosity then directly effects thermal conductivity as more porous regions have lower thermal conductivity. Thermal conductivity has a direct effect on fuel temperatures. Constituent redistribution is worse for high contents of Pu.

5. Thermal conductivity generally decreases quickly with burn-up as the fuel becomes more porous. In particular, high temperature gamma phase regions can see large porous microstructures. Eventually, the thermal conductivity increases again as sodium infiltrates, sodium enters the porous fuel (15-20% of the porosity) as fission gas is released and Na has a higher k than the fission gas. The sodium here is bond sodium in metallic fuel. Swelling due to fission product buildup and transport drives the porosity. 8/8

6. FCCI is fuel cladding chemical interaction. It is a combination of chemical interaction between fuel and cladding leading to weaker cladding (wastage) or local deposits of low melting points. It is dependent on the alloys used. Well known issues are the formation of U_6Fe or Pu_6Fe which may form and have melting temperatures as low as 700K which may put limits on coolant temperatures. The species that lead to the highest cladding wastage due to the formation of a wedge shaped cracks are Lanthanides, Ln and (c). Iron, Ni, and Cr are seen in FCCI from the cladding with Fe and Ni having the highest penetration of the fuel. Lanthanides are the highest cladding reactive. 14/14

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7. Restructuring of mox fuel is mainly driven by temperature and not irradiation. (it can occur rapidly in a cycle). It is due to the thermal growth of grains in the equiaxed grain region. Four regions form; a central void, columnar grain, equiaxed grain, and as fab region. The central void/columnar grain regions form as high-T (1800C) lenticular pores experience a high T delta across the pore causing vaporization of fuel on the hot side of the pore and condensation on the cold side. The pores coalesce into a central void and the fuel grain microstructure is repurposed into large columnar grains following condensation. Differential pressure across the pores also help them moving centrally.

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8. Plutonium concentration is generally higher at the edge of the void in the pin as U is more volatile and restructures towards the edge of the fuel. Additionally, Oxygen moves towards the edge of the fuel. MOX fuel will have higher Oxygen potential until Mo oxidizes which can lead to VO_3 formation that deposits towards the edge of the fuel. This leaves enriched PuO_2 deposits behind. Oxygen also generally follows the thermal gradient downward (vapor/solid diffusion).

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- Kind of mixed/combined some concepts

4. JOG is joint oxide gaine. It is a material that varies in composition but is likely partially made up of volatiles oxidized and precipitated at the fuel/clad gap. JOG can be positive as it provides a mechanical buffer against FCMI, provides better conductivity than gas, and reduces swelling due to fission product release. However it can lead to increased FCCI as volatiles attack Cr in the cladding and lead to clad wastage. In particular, Cs can cause oxide layers to decrease. JOG is formed when volatiles vaporize and migrate to cooler fuel and begin to precipitate with cladding constituents. Combinations of Cs, Mo, and O may form some JOG species. Eg matrix attack and intergranular attack are possible FCCI phenomena for MOX fuel with JOG.

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10. Sodium corrosion occurs with two methods: one is the dissolution of elements and the other is chemical reaction with elements. Dissolution largely occurs due to the thermal gradient in the loop. Solubility in Na increases with temperature and steel constituents dissolve in the high-T part of the loop and condense out on the low-T side of the loop (after H's). Type II corrosion largely occurs due to the presence of impurities. Oxygen, carbon, and Nitrogen can affect this corrosion. In particular, oxygen is much more soluble in sodium than in water. Oxygen may enter the reactor during refueling and can cause Cr/Fe to react more frequently with sodium. Austenitic and Ferritic steels may carburize or decarburize depending on temperatures and Cr concentrations. High-T leads to carburization while High-Cr reduces carburization. Corrosion rate may be affected by other factors like Na velocity, and immersion time as well. Na velocity increases corrosion up to a point where a boundary layer forms. Austenitic steels have an Arrhenius corrosion relation with immersion time but high Ni steels do see increased start-up corrosion while Ferritic steels see steady state corrosion from the start.

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